How the Joux-Vitse cover cost S/rho scales with p (2026-10-24)

Scope. This curve family only: the `jv_cover` genus-3 cover of E(F_{p^6}) over F_{p^2} with Nagao decompositions (full rational 2-torsion, weak class, `generate_spec(p, seed)`), sparse-FGLM linear stage. It is not a statement about prime-order curves. This is a scaling measurement; it makes no speedup claim and no claim to beat rho. Every S/rho below is a lower-bound comparison (AGENTS.md section 5a, see below), reported as the distance remaining.

Labels: **measured** = read from a run record; **derived** = computed from measured values; **model** = cost-model prediction or extrapolation.

Files: `PROTOCOL.md` (frozen before the main run; sha256 `81b2a474...` in `results/FROZEN.sha`, unchanged), `REPORT.pdf`, graph `scaling.svg` / `scaling.png`, raw records and logs in `results/`, fit tool in `fit/` (standalone Rust), `run.sh` (first execution; its 1009 loop was killed in the incident) and `run_1009.sh` (the 1009 runs, thin orchestration).

Result

Pre-registered hypothesis H1: log(S/rho) falls roughly linearly in log p. Cost model: S/rho ~ 720 C_cov / (c_e S_rho p^2), slope **-2** (derived in PROTOCOL.md).

quantity	value	label
fitted slope of ln(S/rho) on ln p, 17 per-seed points, p = 53..1009	**-2.05**	derived from measured
95% interval, t (15 df)	[-2.28, -1.83]	derived
95% interval, cluster bootstrap over seeds (10000 resamples)	[-2.29, -1.84]	derived
half slopes: p <= 151 / p >= 251	-1.60 (se 0.43) / -1.91 (se 0.13); difference -0.31	derived
extrapolated p* where the fitted line meets S/rho = 1	**about 1330**, bootstrap 95% [1170, 1540] (slope-interval ends: [1110, 1660])	**model** (extrapolation)
p* with slope fixed at -2 (cost model, fitted constant)	about 1400	**model**

Decision rules from PROTOCOL.md: the 95% interval lies below 0 and the half-slopes differ by 0.31 < 0.7, so **H1 is supported** on this family; the point estimate -2.05 lies in [-2.4, -1.6] and the interval lies inside [-2.5, -1.5], so the cost model's slope **-2 is consistent and not falsified**; the earlier "about p^-1.2" reading (two seeds at 53 and 101) is **outside** the interval. Caveat: the rule's falsification test only rejects slopes whose interval lies entirely outside [-2.5, -1.5]; here the interval [-2.28, -1.83] lies inside it, so the data neither falsify nor sharply confirm values between.

The largest measured prime, p = 1009, has S/rho between 1.59 and 1.98 (three seeds, **measured**, with rho S taken from the pooled smaller-p runs): the line has not met rho; the measured points do not cross S/rho = 1, so p* is an extrapolation, not an interpolation. Distance remaining at p = 1009: a factor of about 1.7 (geometric mean 1.73).

Per-seed S/rho

S/rho = total counted multiplications / (rho_ref sqrt(l) c_add_E) with the pooled rho reference (**derived**, rho_ref = 1.344, below). "own rho" is the binary's own `s_over_rho` from that run's 16 rho runs (p <= 251 only).

p	seed	S/rho, pooled rho	S/rho, own rho
53	1	544.46	495.8
53	2	244.69	305.66
53	3	1786.8	1476.58
101	1	167.01	156.31
101	2	271.98	243.55
101	3	241.94	244.64
151	1	119.75	108.17
151	2	122.18	146.34
151	3	105.45	117.94
251	1	22.76	26.52
251	2	26.34	25.75
503	1	8.8	n/a
503	2	4.97	n/a
503	3	8.08	n/a
1009	1	1.98	n/a
1009	2	1.59	n/a
1009	3	1.66	n/a

Per-prime geometric means (pooled rho): p = 53: 620 (3 seeds, range 245-1787); 101: 222 (167-272); 151: 116 (105-122); 251: 24.5 (22.8-26.3, 2 seeds, see failure below); 503: 7.07 (4.97-8.80); 1009: 1.73 (1.59-1.98). The seed spread is large at small p (a factor of 7 at p = 53) because relation yield depends on the curve: residuals needed per relation ranged from 8486/33 to 62880/26 at p = 53 (residual / 720|F| from 0.36 to 3.36), and narrows with p (0.98-1.29 at p = 1009).

Run details (measured)

p	seed	|F|	residuals	relations	residual / (720|F|)	incomplete tests	C_cov (F_pmul)	S	LA share (r)	wall s
53	1	30	19020	31	0.85	7	945536	731.6	0.0111	78
53	2	32	8486	33	0.36	5	952378	328.8	0.0173	39
53	3	25	62880	26	3.36	20	938649	2400.9	0.0071	255
101	1	52	40100	53	1.05	4	952058	224.4	0.0046	114

p	seed	|F|	residuals	relations	residual / (720|F|)	incomplete tests	C_cov (F_p mul)	S	LA share (r)	wall s
101	2	49	65422	50	1.82	3	950383	365.5	0.0041	160
101	3	49	58201	50	1.62	3	950313	325.1	0.0046	111
151	1	74	95654	75	1.77	4	956408	160.9	0.0027	176
151	2	70	97696	71	1.91	2	955355	164.2	0.0032	246
151	3	70	84292	71	1.65	3	955662	141.7	0.0031	210
251	1	129	82263	130	0.88	1	970514	30.58	0.0023	191
251	2	123	95284	124	1.07	1	969913	35.39	0.0017	375
503	1	238	252255	239	1.47	1	985463	11.83	0.0008	551
503	2	262	142150	263	0.75	1	986447	6.67	0.0010	313
503	3	249	231408	250	1.29	2	985735	10.85	0.0009	512
1009	1	484	450841	485	1.29	1	1000926	2.66	0.0004	620
1009	2	511	360835	511	0.98	0	1001264	2.13	0.0005	494
1009	3	497	377514	498	1.05	0	1001006	2.23	0.0004	558

C_cov is nearly flat in p (9.4e5 to 1.0e6, +6% over a factor 19 in p) and |F| is about p/2, as the model assumes; so the slope is dominated by the p^-2 from sqrt(l) ~ p^3/2 against relations ~ p. Linear algebra is at most 1.7% of S (consistent with "relation collection ~99.99%" only at large p; at p = 53 it is up to 1.7%). Wall times are practicality notes on a shared 4-vCPU host, not the metric.

Rho reference

Pooled mean of 176 rho S samples at p <= 251 (the `rho_e` walk, every group operation counted): **1.344** (single-run sd 0.62, se 0.047; **measured**). Per prime: 53: 1.39, 101: 1.42, 151: 1.27, 251: 1.26 (48, 48, 48, 32 runs). The trend of rho S against ln p is -0.10 per unit ln p, 95% interval [-0.27, 0.07]: no significant trend, so pooling is not rejected, but the interval is wide and the point estimate is a decrease of 0.1 in rho S per unit ln p. **The 503 and 1009 rows use this pooled value as a model assumption**: rho cannot be run there (l ~ 2^52 and 2^58, above the 2^48 memory limit of the stored-table rho). If rho S really kept falling at the point-estimate trend (not significant), rho S at 1009 would be about 1.12 and S/rho there about 20% higher than shown. The fit with each point's own rho (p <= 251 only, 11 points) gives slope -1.90, 95% [-2.48, -1.32], p* about 1640 (**model**); it is consistent with the primary fit and less precise.

Controls (measured)

- Every `dlp` run that finished: `solved = true`, `correct = true` (logarithm equals the planted secret), `mismatches = 0` (17 of 17 runs; oracle cross-check ran on 1414 residuals at p = 53 only, because the oracle table needs |F| <= 40), every relation verified by `verify_dec` before use.

- `ccov` control per prime (60 tests each): planted decompositions found 20/20 at every p in {53, 101, 151, 251, 503, 1009}; oracle mismatches 0; unverified 0; incomplete 0. The oracle itself checked 60 residuals at p = 53 and none at larger p (|F| > 40 in this binary's setting), so the oracle control only covers p = 53.
- Determinism control: a second full execution (after the incident below) reproduced `total_muls` bit-for-bit in 11 of 11 runs that it completed (p = 53 to 251, `results/replicate/`).
- Incomplete tests (a Nagao test whose Groebner stage did not finish) occurred in every run (0 to 20 per run); they lower the relation yield, are counted in S, and cannot produce wrong relations.

Section 5a pricing band

Not applicable as a ranking, and stated precisely: the binary's counted unit is F_p multiplications priced into group additions. For rho the unpriced native work is a hash, a table lookup and a table insert per step (3 per step, derived from `rho_e`); for the cover arm the unpriced F4 additions and reductions, memory traffic and Wiedemann vector work are **not emitted** by the binary. An arm that does not emit its counts cannot be ranked under section 5a, so this is recorded as an implementation gap and every S/rho here is a lower-bound comparison at weight 0 on both arms. Sensitivity only (rho side repriced at w = 0, 0.1, 1 GAE per native operation, rho cost x (1 + 3w), cover arm left at 0, **not a valid band**): the p* extrapolation would move from about 1330 to 1170 and 680. The slope is unchanged by any p-independent w. No cover-versus-rho ranking is made.

Failures and deviations (recorded, not worked around)

1. **p = 251, seed 3: killed by the out-of-memory killer** (exit 137, cgroup memory limit, `jv_cover` RSS 13.9 GB) in the 16-run stored-table rho phase, after 324 s. No JSON was written; the log is empty. Per PROTOCOL.md failures are reported and not silently replaced, so p = 251 has 2 seeds, not 3. The measurement itself is unaffected (rho memory, not the cover).
2. **Working-tree incident (caused by the parent session, not by the experiment).** At about 05:50Z the study files disappeared when the parent session's git checkout switched branches after snapshotting the study to PR #2484's branch. The parent restored every file as untracked afterwards. Meanwhile this session, not knowing that, re-executed the study from a scratchpad copy (frozen PROTOCOL.md recreated byte-identically: sha256 matched) and, on the restoration notice, stopped that second execution. Its 11 completed runs are the determinism control above; the first-execution records are the data. The p = 1009 runs were made after the restoration with `run_1009.sh` in the study's `results/` (seeds 1, 2, 3, all finished, 494-620 s each; the 25-minute gate passed). `results/walls.txt` has the first execution's lines; the 1009 lines were appended afterwards. The interim snapshot in `reports/interim-20261024T0550Z/` was produced by the parent session before the 1009 runs, and uses Python for plotting; it is not used here.
3. `convert` could not render the SVG (no `rsvg-convert`); the PNG was rendered with LibreOffice headless from the Rust-written SVG. The PDF was built with pandoc to ODT

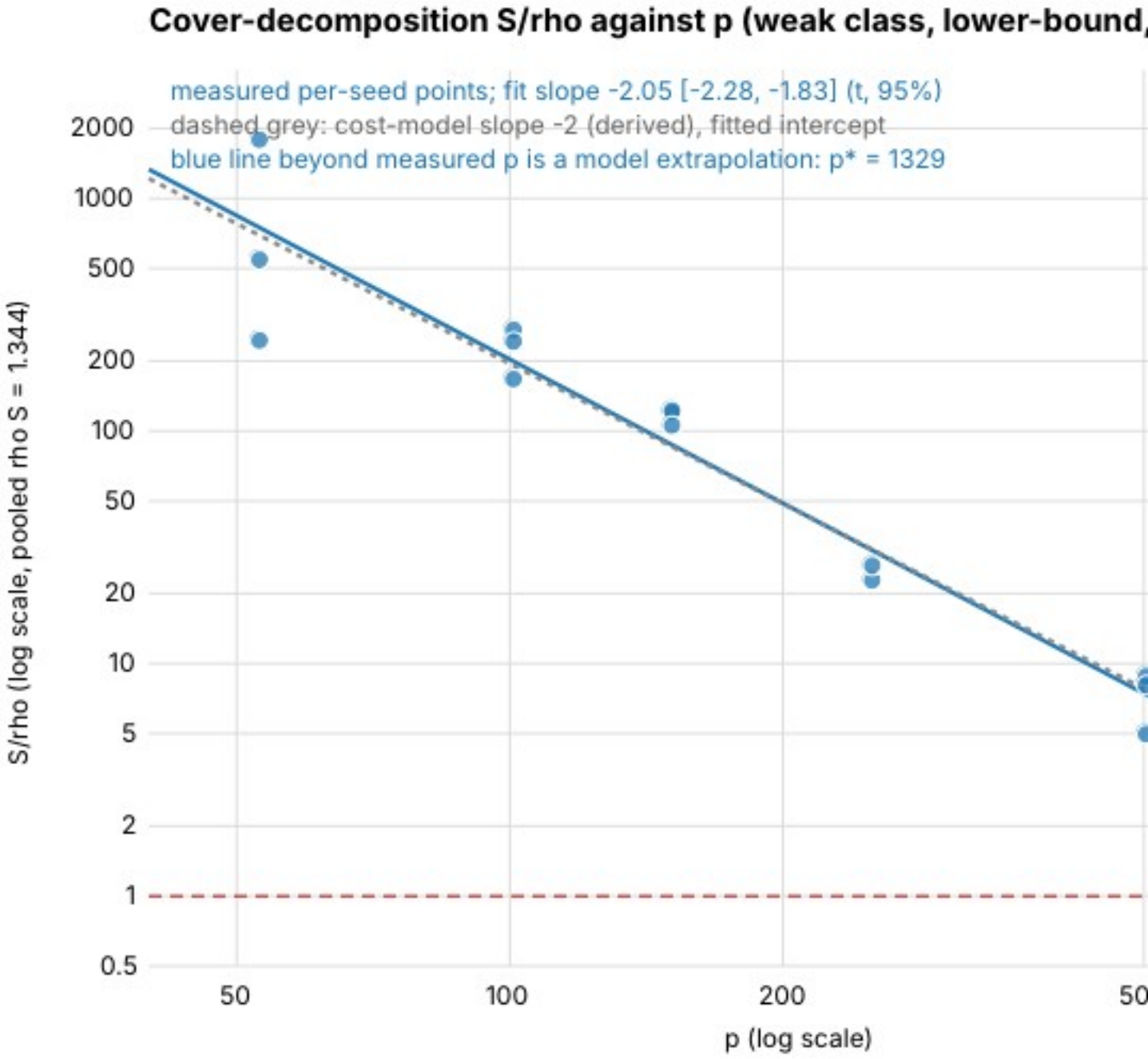
and LibreOffice headless; no TeX or typst engine is installed.

4. The rho reference at p = 503 and 1009 is a pooled model value, not a measurement (above).
5. The binary was run with `--rho-ref 1.3` at 503 and 1009 only so it would finish; the `s_over_rho` printed there is not used.

Reproduce

```
cargo build --release --example jv_cover
research/jv_cover_p_scaling_20261024/run.sh;
research/jv_cover_p_scaling_20261024/run_1009.sh
cd research/jv_cover_p_scaling_20261024/fit && cargo build --release --
offline && ./target/release/jv_p_scaling_fit ../results
```

Binary sha256 `a045cb80...e34d` (`results/versions.txt`, `results/probe/bin.sha`).


Cover-decomposition S/rho against p (weak class, lower-bound,
measured per-seed points; fit slope -2.05 [-2.28, -1.83] (t, 95%)
dashed grey: cost-model slope -2 (derived), fitted intercept
blue line beyond measured p is a model extrapolation: p* = 1329
S/rho (log scale, pooled rho S = 1.344)
2000
1000
500
200
100
50
20
10
5
2
1
0.5
50
100
200
50
p (log scale)